Lab Exercise 7

Objectives:

At the end of the lab exercise, student should be able to:

- **Understand the concept of functions without parameters**
  - Students will learn how to define and call functions that do not require input arguments, reinforcing the idea of modular programming.
- **Develop structured problem-solving skills through modular design**
  - By separating tasks into input, processing, and output functions, students will strengthen their ability to design programs systematically and improve code readability.

Task 1 : Basic Function

Write a C program that defines **three functions without parameters**. Each function should perform a simple task:

1. The first function should print a welcome message.
2. The second function should perform a basic arithmetic operation (for example, add two fixed numbers) and display the result.
3. The third function should print a goodbye message.

Call all three functions in sequence inside the **`main()`** function.

Task 2: Apply function with selection and looping statement

Write a C program that performs basic arithmetic operations using functions. Your program must:

1. **Display the following menu** repeatedly until the user chooses to exit:

   ```
   1. Add
   2. Subtract
   3. Multiply
   4. Division
   0. Exit
   ```

2. Create **a function to display the menu**.
3. Create **separate functions** for each operation:
   - `add()`
   - `subtract()`
   - `multiply()`
   - `division()`

4. The program must:
    - Continuously show the menu inside a loop.
    - Ask the user to enter an option.
    - Call the appropriate function based on the user's choice.
    - Exit only when the user presses **0**.
5. Each operation function should:
    - Accept two numbers as input.
    - Return the result to the main program for display.

Task:
Implement the complete program using functions, loops, and conditional statements.